

SensAiry: A Health Tracker for Cars



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SensAiry solves a problem that might seem simple at a superficial level but eventually ends up causing a significant amount of trouble to millions of people. If you were looking at an elevator pitch for SensAiry, it could be this: "It enables people to check their vehicles' tyre pressure and temperature using a smartphone."

Prabu Surendra, CEO, Tymtix, whose team built SensAiry, says that research has shown that vehicles with under-inflated tyres are three times more likely to be involved in a road accident. "Yet, people do not check the tyre pressure as often as they should. One reason could be to avoid the hassle of going to a service station and waiting in queue to get the tyre pressure back up to the recommended manufacturer specified levels," he says.

Petrol-heads could be early adopters

If you are a car or bike enthusiast, you will know how important it is to maintain the optimal tyre pressure and tempera-

ture to maximise traction. SensAiry eyes and ears, so to say, are a temperature and pressure sensor packed together. The rest of the device focuses on gathering data and comparing it to set thresholds programmed into the firmware to make the decision as to whether there is a problem that the user should know about.

The device uses a coin-cell battery and works on a 2.4GHz frequency, which enables a longer coverage for

the device while measuring up to 150psi at almost 10,000rpm.

Under normal conditions, it collects data and transfers it to the smartphone running SensAiry app, where the user can go through the information presented on it. However, if there is a situation involving, say, a heated-up tyre or a puncture, the device sends out notifications to the driver so that she or he can do what is necessary.

The best part is that, what comes after the device gathers a lot of data. "For example, let's say there is a very small puncture on a car tyre. Users would not be able to see the tyre pressure variance over a period of a few days, but the app collects data and can alert vehicle owners to these minor punctures that are not obvious to them," explains Surendra. The product also lets you track multiple vehicles simultaneously.

Essentially, what they have done here is data visualisation and analytics on tyre and pressure data. It is somewhat similar to what the billion-dollar company Jawbone has famously done with their UP series of bands.

Another element that will be included in future models is shock detection. An accelerometer that senses shocks caused due to bumps and road problems will help users better estimate and even elongate the life of the tyre by making sure they are taking care by keeping the tyre properly inflated. It will also help them properly maintain the tyre pressure during dry and rainy months when it is recommended to slightly decrease or increase the pressure to maximise performance.

The accelerometer can also help users judge the performance of the suspension setup. If they have spent a bomb on getting Koni, Bilstein or Tien shocks for their cars, they can use this app to follow up on the installation to track how shock-data changes for the car while running on the same strip before and after suspension upgrade.

What Surendra is more excited about is the potential to use the data flowing in

SensAiry device

